



Inspiring Excellence

Department of Electrical and Electronic Engineering
Hardware / Software Report Submission

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Course Title: Electrical Circuits I Laboratory

Section: 03

Experiment No: 6+7

Experiment Name: Generation and Measurement of Waveforms, Study of Transient Behavior of RC Circuit

Group No: 01

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Electrical Circuits I Laboratory

EEE 101L

Department of Electrical & Electronic Engineering (EEE)

Brac University

Experiment No. 6+7

Generation and Measurement of Waveforms, Study of Transient Behavior of RC Circuit

1. Objective:

- To understand the operation of an AC voltage source (function generator) and an oscilloscope.
- To analyze the exponential charging and discharging behavior (transient response) of a series RC circuit under a square wave step input.

2. Theoretical Background:

An RC series circuit consists of a resistor R and a capacitor C connected in series with a voltage source.

Basics and Mathematical Expressions

When a sudden step voltage V is applied (charging phase), the voltage across the capacitor (V_C) and resistor (V_R) vary exponentially over time t based on the time constant $\tau = RC$:

$$V_C(t) = V \left(1 - e^{-\frac{t}{\tau}} \right)$$

$$V_R(t) = V e^{-\frac{t}{\tau}}$$

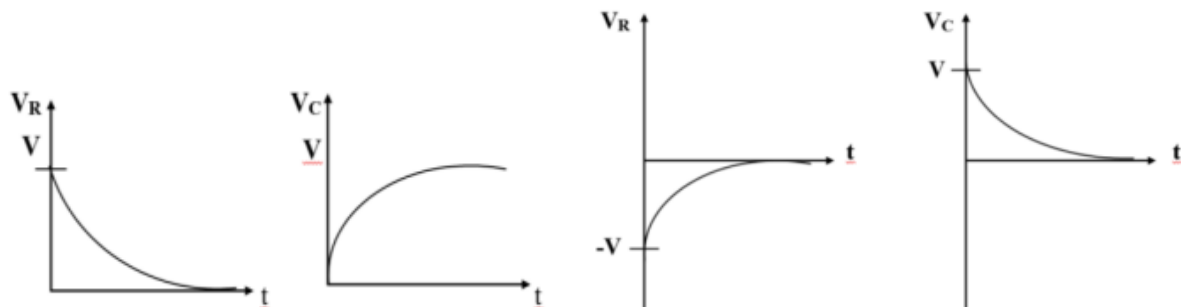
When the source drops back to zero volts (discharging phase), the stored energy discharges through the resistor:

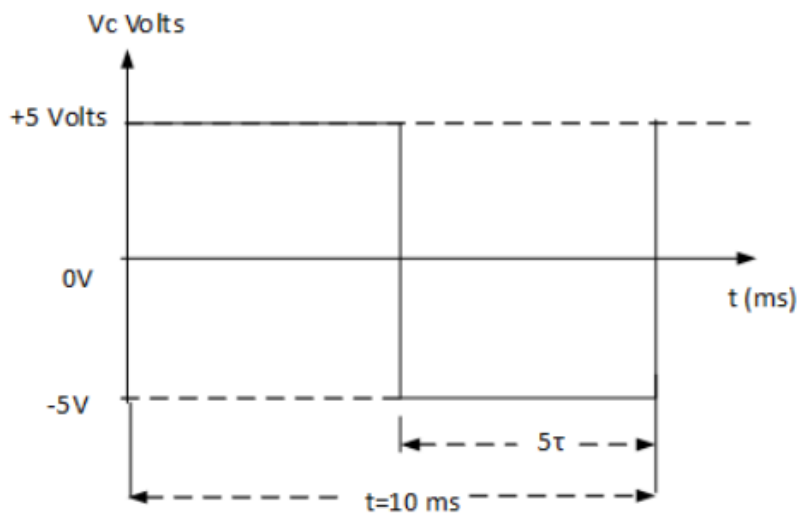
$$V_C(t) = Ve^{-\frac{t}{\tau}}$$

$$V_R(t) = -Ve^{-\frac{t}{\tau}}$$

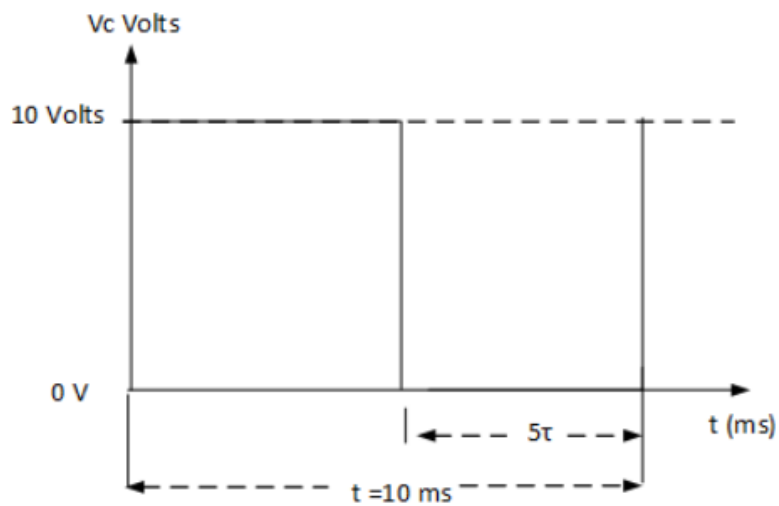
Input/Output Waveforms Relation

- **Input Waveform (V_{in}):** A continuous square wave that alternates between a high voltage state (acting as a charging step input) and a 0 V ground reference state (acting as a discharging path).
- **Capacitor Output (V_C):** An exponential curve that rises toward the maximum input voltage during the high state and decays exponentially back to 0 V during the low state.
- **Resistor Output (V_R):** Sharp positive and negative alternating exponential spikes that rapidly decay back toward zero at every sudden transition of the input square wave.

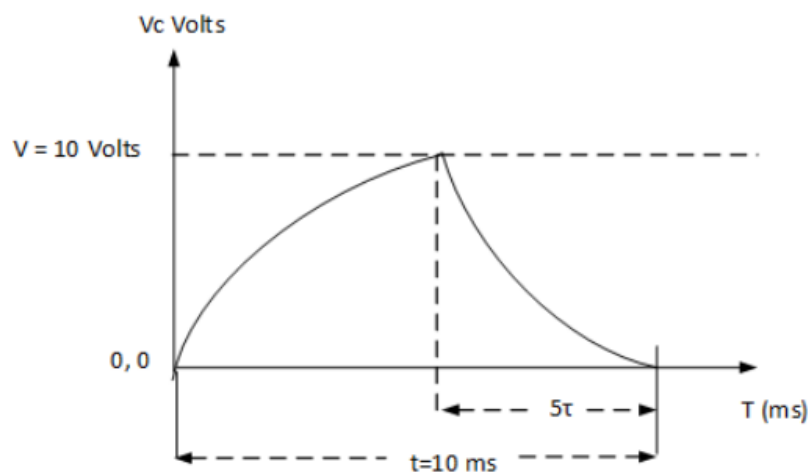




An NRZ Square wave with frequency of 100 Hz with amplitude +5 Volts
- 5 Volts volts along the reference axis
 $\tau = 1 \text{ ms}$



Square wave = 100 Hz with amplitude 10 volts offset adjusted above the reference axis

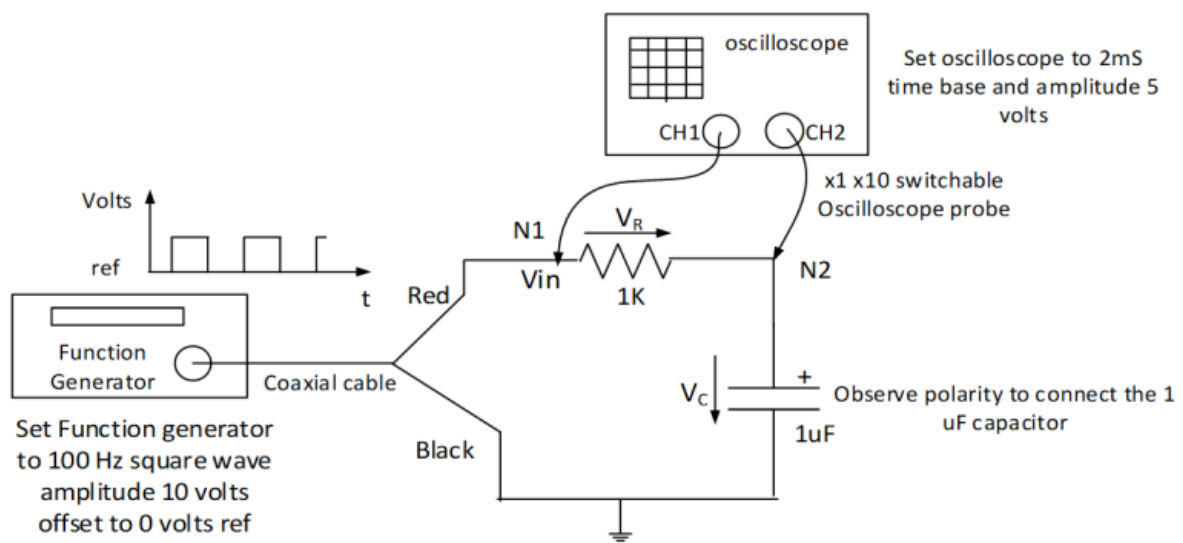


Waveform showing charging and discharging of an RC circuit

3. Equipments:

- Resistor: $1\text{ k}\Omega$
- Capacitor: $1\text{ }\mu\text{F}$
- Oscilloscope and Probes
- Signal Generator (Function Generator) and Cables
- Wires
- Breadboard

4. Circuit Diagram:



Experimental setup to observe transient
behaviour of a RC circuit

5. Results:

Theoretical Time Constant (τ) Calculation

Using the nominal values given in the lab experiment setup:

- $R = 1 \text{ k}\Omega = 1 \times 10^3 \Omega$
- $C = 1 \mu\text{F} = 1 \times 10^{-6} \text{ F}$

$$\tau = R \cdot C$$

$$\tau = (1 \times 10^3 \Omega) \times (1 \times 10^{-6} \text{ F}) = 1 \times 10^{-3} \text{ s} = 1 \text{ ms}$$

The theoretical time constant of the circuit is exactly 1 ms.

6. Images taken from oscilloscope:

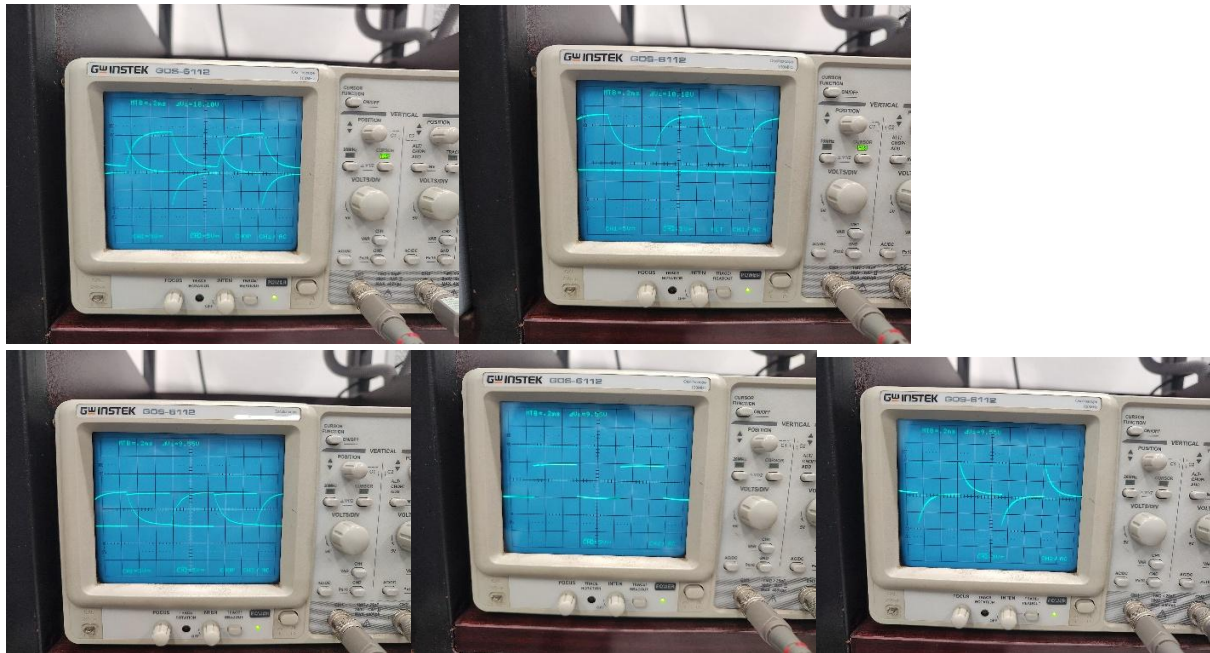


Fig (In Sequence): V_C & V_R overlaid, V_C waveform, V_{in} & V_C overlaid, V_{in} waveform, V_R waveform.

7. Conclusion / Discussion:

Question 1: Define time constant for an RC circuit.

Answer: The time constant ($\tau = RC$) is the time period required for the capacitor voltage to charge up to 63.2% of its maximum steady-state value, or to discharge down to 36.8% of its initial stored voltage.

Question 2: What is the significance of the time constant?

Answer: It serves as a direct indicator of circuit latency, establishing how fast or slow a passive network structurally transitions and stabilizes when exposed to a sudden change in input voltage.

Question 3: How can the time constant be determined?

Answer: Theoretically, it is calculated multiplying component values ($\tau = RC$) [cite: 1]. Experimentally, it is found from the oscilloscope trace by measuring the time elapsed for the capacitor voltage (V_C) curve to scale from 0 V to 6.32 V when a 10 V input step is active.

Experiments 6 and 7 successfully demonstrated how to operate an oscilloscope and an AC function generator to observe and analyze the transient response of a series RC circuit under a 100 Hz square wave input. The observed charging and discharging voltage waveforms on the oscilloscope perfectly matched the theoretical exponential transitions predicted by mathematical circuit models.